

Subject: Digital Communication [EC8551]
B.Tech III (Sem 5) Mid Sem Exam- October 2023

Time: 1.5 Hrs

Attempt any 5 Questions: Each question carries 6 marks.

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Question number	Question																		
Q-1	A 7 bit PCM system employing uniform quantization has an overall signaling rate of 56 kbps. What is the theoretical maximum frequency (f_m (bandwidth)) that this system can handle? Based on channel SNR 35 dB what is the channel capacity for error free transmission? If this SNR is dropped by 50% what will be the % drop in channel capacity [Hint: signaling rate r = number of bits n $\times$ sampling rate f_s]	[6]	CO4,5,6																
Q-2	(a) What is the importance of Kraft inequality condition? (b) Draw the model of Binary Symmetric Channel. (c) An instantaneous adaptive delta modulator adapts its step size exponentially. i.e. if the predictor is unable to track the input, the step size is doubled. Thus the steps generated have the values given by $S_n = 2^n$, where $n = 1, 2, 3, \dots, 10$. Find the Entropy of the step generator if the probability of a step S_n is $p(S_n) = 1/S_n$ for $n = 1, 2, 3, \dots, 9$ and $p(S_{10}) = p(S_9)$.	[1.5] [1.5] [3]	CO1 CO1 CO3																
Q-3	A DMS X has six symbols x_1, x_2, x_3, x_4, x_5 and x_6 with $p(x_1) = 0.3, p(x_2) = 0.19, p(x_3) = 0.16, p(x_4) = 0.15, p(x_5) = 0.1$ and $p(x_6) = 0.1$. (a) Construct a Shannon-Fano code for X and calculate the efficiency of the code. (b) Repeat for the Huffman coding and compare the results.	[6]	CO4,5,6																
Q-4	(a) If VLC code table is as follows, read and write the message for the given pattern from the LSB side 111000010001010101110101 <table border="1"><thead><tr><th>Message</th><th>Code</th></tr></thead><tbody><tr><td>M1</td><td>1</td></tr><tr><td>M2</td><td>01</td></tr><tr><td>M3</td><td>001</td></tr><tr><td>M4</td><td>0000</td></tr><tr><td>M5</td><td>00010</td></tr><tr><td>M6</td><td>000110</td></tr><tr><td>M7</td><td>000111</td></tr></tbody></table> (b) Explain in detail the error control mechanisms in each case with proper timing diagram.	Message	Code	M1	1	M2	01	M3	001	M4	0000	M5	00010	M6	000110	M7	000111	[2] [3]	CO3 CO2
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